

Jialin Huang

Backend Engineer

jialin00@gmail.com [LinkedIn](#) [GitHub](#) jialin00.com

Backend engineer with a frontend foundation, building REST and GraphQL APIs on a multi-tenant cybersecurity SaaS: policy config, RBAC, SAML, Cognito audit. On my own I go deeper: a Go URL shortener that splits OLTP from OLAP, and a RAG pipeline with its own eval harness. I build APIs the client actually wants to consume.

EDUCATION **National Chengchi University (NCCU)** · B.A. in Economics

EXPERIENCE

MammothCyber Senior Full Stack Engineer

Apr 2025 - Present · Cybersecurity · Taipei (Hybrid)

Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; security domain (access control, audit, policy config)

NESTJS APOLLO GRAPHQL GOLANG GIN POSTGRESQL DOCKER KUBERNETES JENKINS AWS LAMBDA AWS COGNITO

Policy Configuration Console

- Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request

Secure Browser & Extension (Chromium)

- Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Identity, Access & Audit

- Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients
- Integrated Google / Apple social login into Cognito on top of the existing SAML, reconciling the OAuth scopes for external IdPs and reworking several Cognito Lambda triggers, including the PreTokenGeneration hook that maps each social user to its tenant
- Integrated credentials and storage across AWS, Azure, and GCP (presigned URLs, Key Vault signing)

Platform & Build Infrastructure

- Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0
- **npm to pnpm**: Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree
- Cut two Docker images ~80% with bundling and pnpm deploy --prod: Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB; compressed registry size ~540 MB → 89 MB (-84%) and 628 MB → 180 MB (-71%)

Bunastar Frontend Engineer

Nov 2024 - Apr 2025 · Gaming · Taipei (Hybrid)

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

WEBSOCKET GOOGLE ANALYTICS

Southeast Asian Gaming Platform - Pok Deng

- Real-time card sync over WebSocket

Confetti Full Stack Engineer

Dec 2022 - Oct 2024 · Gaming · Taipei (Hybrid)

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SCMP Frontend Engineer

Oct 2021 - Nov 2022 · Media · Hong Kong (Remote)

South China Morning Post: high-traffic international news platform, delivered fully remote

NODE.JS **GRAPHQL**

scmp.com ↗

- Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)

Futuresai Frontend Engineer

Nov 2020 - Oct 2021 · Finance · Taipei (Hybrid)

SIGNALR **FIREBASE** **AUTH** **FIREBASE**

Trading Dashboard + Admin + Auth

- Monorepo rewrite with SignalR for real-time trading-data streaming
- SMS and LINE push notifications
- Firebase auth middleware with custom membership logic

Dentalltw Frontend Engineer

Jul 2018 - Nov 2020 · Healthcare · Taipei (Hybrid)

Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market

PYTHON/FLASK **DOCKER** **GCP**

CI/CD Pipeline

- Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

SAVY Frontend Intern

Apr 2018 - Jul 2018 · Taipei (Hybrid)

N/A FOR THIS VIEW

PROJECTS

URL Shortener ↗

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

GO **GIN** **POSTGRES****SQL** **REDIS** **KAFKA** **CLICKHOUSE**

- Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect
- Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta
- Live click dashboard over Server-Sent Events: a pub/sub hub fans out each redirect to all clients, plus a per-link 2s stream merging the Postgres total with the Redis delta
- Hand-rolled a Redis-bitset Bloom filter to short-circuit lookups for keys that do not exist

221b

My outie and my innie in one searchable self. Severance, but merged instead of split. Ask what my brain has shipped, how I think, or "what would I do?", and it answers as me.

PYTHON **CHROMADB** **SENTENCE-TRANSFORMERS** **MCP**

- One corpus of my blog, my Claude sessions, and my inner config (soul, persona, style).
- Chinese question → embeddings retrieve → reranker reorders → my English writing.
- Answers "if you were me, how would you decide?" from my values and past calls.

WRITING

- [N+1 isn't a database problem. It's a shape.](#) ↗
- [HTTP 1.1, 2, and 3: what broke and what fixed it](#) ↗
- [Five ways to reach a service in Kubernetes](#) ↗
- [A packet's walk through a VPC, in AWS](#) ↗
- [Stopping XSS and CSRF: who's responsible for what](#) ↗

